

# A New Deep Reinforcement Learning-Based Adaptive Traffic Light Control Approach for Isolated Intersection

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**Abstract**— In this work, we focus on optimizing traffic signal control at an isolated intersection and subsequently alleviate the traffic flow. We propose a new Deep Reinforcement Learning-based approach. Thus, the traffic network controller in an isolated intersection is modelled as an intelligent agent that perceives the discrete state encoding of traffic information as the network inputs. Our contribution resides to use a Double Deep Q-Network (DDQN). This argues that the idea of having a simplified state and reward formula facilitates the training of the agent by simplifying the convergence of the latter. It dynamically select the phases improving the traffic quality.

The experimental results shows that the proposed approach is competitive in terms of Average Waiting Time, Average Queue Length, Average Fuel Consumption and Average Emission CO<sub>2</sub> at intersection when compared to some baseline methods.

**Keywords**— reinforcement learning; traffic light control; deep reinforcement learning; adaptive traffic signal control; SUMO.

## I. INTRODUCTION

Traffic congestion problem is a critical phenomenon that affects the majority of cities' environments around the world. The amelioration of infrastructure has been the primary solution to face such problem. Nevertheless, this is not often always possible, notably when the financial constraints are too tight. As a result, various alternative approaches have been considered, which are often centred on improving the efficiency of existing infrastructure by using traffic lights control systems at intersections. At the time, the traffic lights are controlled independently of the real traffic situation with a predefined fixed-time plan [1]. Generally, this kind of methods is not optimal, since it ignores the present traffic condition, resulting in long queues and waiting times.

With the significant advancement in the Artificial Intelligence (AI) and the Internet of Things (IoT) areas, many Intelligent Transportation Systems (ITS) have been proposed to improve traffic network quality based on the real-time traffic state [2]. In this perspective, adaptive traffic lights control based-approaches have been considered as the most used and has shown to be a favourite methods to relieve traffic congestion [3].

A hopeful approach consists to use Reinforcement Learning (RL) techniques. These last allow the traffic lights control system to automatically learn a pertinent or even optimal policy by using interactions with the environment, while maximizing a reward (R) function. Many DRL-based

approaches have been proposed proving the interest of scientific and economic communities to enjoy the benefits of these techniques to improve the performance of traffic networks [4]. Such approaches have tried to use RL techniques to solve the problem from different angles and have shown better performance than other methods based on learning [5–9].

This paper represents a new contribution in the field of traffic light control. We particularly focus our research on an isolated intersection controlled by traffic lights. Thus, a new DRL-approach is proposed to dynamically control the traffic network in an isolated intersection. Our problematic is modelled as an autonomous intelligent agent that observes the current traffic state to decide the next green light phase. The consequence of such decision appears on the intersection as a new traffic state and a reward that are given to the agent after the decision is applied. In this way, the controller gets the aptitude to automatically seek the optimal policy according to different states through a training process.

By using the SUMO (Simulation of Urban Mobility) platform, many experiments are carried out to investigate the performance of our proposition. The obtained results indicate that the proposed method outperformed all other approaches that are: DQTSICA (Deep Q-network Traffic Signal Control Agent) [5], the fixed model and TSTMA (Traffic Signals Time Manipulation Algorithm) [17] in terms of Average Waiting Time (AWT), Average Queue Length (AQL), Average Fuel Consumption (AFC) and Average Emission CO<sub>2</sub> (AEC).

The remainder of this paper is organized as follows: Section 2 deals with literature. Section 3 formally defines the problem. The method is presented in section 4 and the experimental results are presented in section 5. Finally, the conclusion of this paper with future direction is summarized in section 6.

## II. RELATED WORKS

RL method has demonstrated its efficiency in dynamic models as a structure for computational neuroscience. The adoption of such method has extensively been studied in ATSC area. Common methods of RL are Q-learning, Sarsa and Policy Gradients. Among them, Q-learning was the primary RL algorithm applied at a single intersection. Indeed, this algorithm has attracted large attention and a numerous researchers focalizing their studies on this area putting in evidence the increasing interest shown by the scientific community to face the associated problematic [4, 7, 10, 11]. In both research works [12] and [13], the authors use a Q-learning strategy with several objective functions to achieve

optimum adaptive signal control for an isolated intersection. Chu et al. proposed in the same context a new mechanism that is based on Q-Learning to optimize Average Vehicle Delay Time (*AVDT*) in an isolated intersection [14]. Moreover, to optimize the Average Delay Time (*ADT*) of the traffic, a Q-learning-based method is developed in [15] for controlling signal lights in an isolated intersection. By combining the Recurrent Neural Network (*RNN*) with Deep Q-Network (named *DRQN*). Zeng et al. propose a new method that is compared with standard Deep Q-Network (*DQN*) [6]. In [18] the double DQN (*DDQN*) is proposed for isolated intersection with duel agent architecture for traffic signal control.

In [16], the authors explored advantage actor critic with different types of status information. Because of the adaptability issue, most centralized *RL* researches have carried out tests in isolated intersections or small traffic networks. A DRL-based cooperative approach is proposed in [19] for multi-intersection traffic signal control.

This research work proposes a new DRL-based approach in this research that optimizes four important metrics (*AWT*, *AQL*, *AFC* and *AEC*) for an isolated intersection controlled by lights. This approach is particularly based on both Double Deep Q-Network (*DDQN*) method and IoT model to control in real-time the traffic network. Therefore, we adopt a scenario including various phases, which corresponds to a large action space in a cycle and ensures that the time of the traffic signal may be changed efficiently from action to other.

### III. PROBLEM DEFINITION AND OBJECTIVES

Our problematic can be described as a model integrating an environment (noted *E*) including an isolated intersection composed of three roads and an intelligent agent (noted *G*) managing the traffic flow in this intersection using a tricolor lights (Red, Green and Yellow). The letters *N*, *S*, *W*, and *E* denote the *north*, *south*, *west* and *east* directions, respectively. For each direction, we consider three lanes indicating all possible senses. In each lane, two sensors (noted *Sensor1* and *Sensor2*) are placed in both extremities to account the real-time number of vehicles.

Moreover, a traffic light setting is defined by a phase. There is a *Yellow* light of 4 seconds when a light changes from *Green* to *Red*, while the other directions remain *Red*. Our problem can be simplified by considering only eight phases. In most cases, as shown in Fig. 1, our scenario is defined by three important parameters: traffic, track entry and phase definition.

The aim of this study is to optimize the congestion in an isolated intersection by using a new DRL-based approach as will well detailed in the next section. Thus, the state information accessible by the agent included all queue lengths of the different lanes and the elapsed phase time.

Numerous machine-learning problems may be affected by the dimensionality problem, deep neural networks may generalize from what they've learnt, and as a result, the challenges caused by dimensional miscalculation are reduced. Additionally, one of the benefits of Rectified Linear Unit (*ReLU*) ability to train the *NN* many times more quickly without significantly reducing the generalization's precision is what prompted us to use a function like this in the approach we proposed.

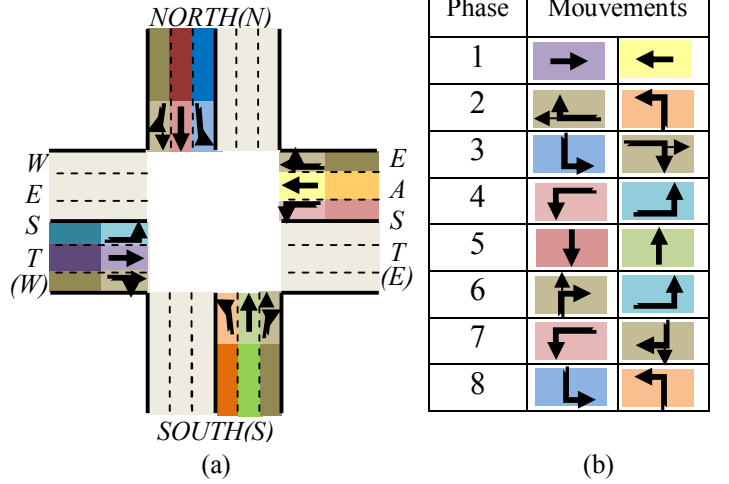


Fig. 1. Our model based on an isolated intersection with 8 phases.

### IV. PROPOSED DEEP REINFORCEMENT LEARNING MODEL

We adopt a DRL-based model through which an agent interact continually with an environment under uncertainty to learn a policy (noted  $\pi(s)/s$  is a state) and achieves a goal. Such agent tries to develop an optimal policy (noted  $\pi^*$ ) that maximizes the anticipated cumulative reward (noted  $R_t$ ).  $R_t$  is defined as:

$$R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k} \quad (1)$$

Where  $\gamma$  represents a discount factor defining the significance of anticipated rewards.

Furthermore, we consider a reward function defined as a change in the cumulative waiting time of the vehicle (noted  $W$ ) and the emission of  $\text{CO}_2$  (noted  $Ec$ ) between two next steps. The equation 2 describes the formula of such function:

$$\text{Reward} = W_t - W_{t+1} + Ec_t - Ec_{t+1} \quad (2)$$

Where  $W_t$ ,  $W_{t+1}$  et  $Ec_t$ ,  $Ec_{t+1}$  represent the accumulated total waiting time and Emission  $\text{CO}_2$  of all the cars in the intersection.

These parameters can be defined as:

$$W_t = \sum_{v \in V_t} w_t^v \quad (3)$$

$$Ec_t = \sum_{v \in V_t} Ec_t^v \quad (4)$$

Where  $V_t$  is the set of vehicles on approaching lanes within the simulation at time  $t$  and  $w_t^v$  and  $Ec_t^v$  is that the waiting time and emission  $\text{CO}_2$  of vehicle  $v$  at time  $t$  respectively.

Recalled that the proposed approach is based on *DDQN* method, which is converges through execution of recurrent actions until achieving a maximum anticipated reward.

Therefore, an updating function of Q-value (noted  $Q(s_t, a_t)$ ) constitute a key element of this method. It is formulated as:

$$Q_{t+1}(s_t, a_t) = r_t + \gamma Q_t(s_{t+1}, \arg \max_{a'} Q_t(s_{t+1}, a', \theta_i)) \quad (5)$$

Where  $\theta_i$  is the evaluating network parameters at the iteration  $i$ . and  $r_t$  is the immediate reward. The DQN target's maximal Q- value is substituted with an action based on the evaluation networks' maximal Q-value. As a result, the overestimation of the Q-value is no longer an issue.

As shown in Fig. 2, the agent implementing our model controls the different lights with a fully-connected DRL in order to calculate the Q-value for state-action pairs at the

intersection. The agent observes the state of the intersection at each step and select the corresponding action, while applying the  $\epsilon$ -greedy strategy the value of  $\epsilon$  is defined as:

$$\epsilon = 1 - \frac{\text{epoch\_current}}{\text{epochs}} \quad (6)$$

where  $\text{epoch\_current}$  is the current episode and  $\text{epochs}$  is the total number of episode.

The  $\epsilon$ -greedy strategy is presented in pseudo-code described in Algorithm 1.

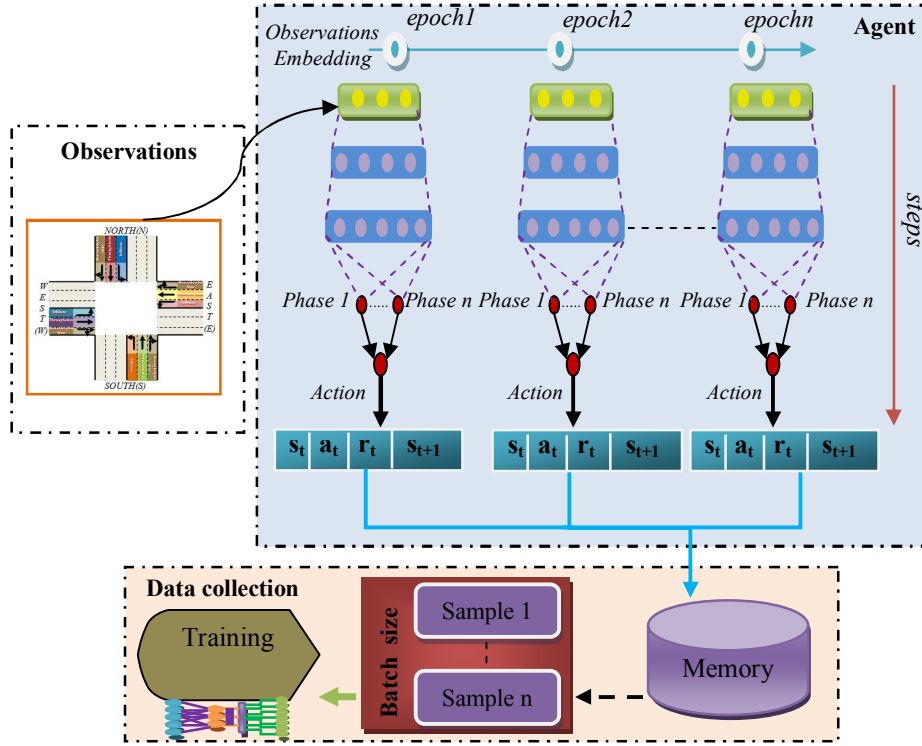


Fig. 2. Structure of proposed model.

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**Algorithm 1:**  $\epsilon$ -greedy strategy

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With probability  $1 - \epsilon$  // exploitation

Choose  $\arg \max_a Q(s, a)$

With probability  $\epsilon$  // exploration

Select random action

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In the proposed approach, the agent selects an action based on its reward. It always selects the optimal action that is the correct phase of traffic lights at an isolated intersection to minimize four metrics: *AWT*, *AQL*, *AFC* and *AEC*. Indeed, it is very important to determine the indispensable components for the learning process. The main of these components are: state, action, and reward. They are defined according to our DRL-based model as follows:

First, as described above, the Eq. (2) defines the reward function. To define the state ( $S_t$ ) at a step  $t$ , all lanes are discretized in cells and all cells' queue length are used as a state to control phase time. Finally, as an action, at the end of each step, the agent selects the next phase to apply to the intersection. All possible actions are well presented in Fig. 1.

Moreover, the training process is considered as a very

important operation of the proposed approach. Thus, as presented in the Algorithm 2's pseudo-code, the agent is trained using a replay process experience that is stored in a memory (noted  $M$ ). From this last, the agent takes samples to use them for updating the network. The memory  $M$  is regularly updated during the training process.

The deep neural network is a fully connected consists of 3 hidden layers, each containing 300 nodes each of which is modified by linear unit activator function (*ReLU*).

## V. EXPERIMENT

The investigation of our model is done using simulation. Indeed, the proposed model is implemented via *SUMO* environment under a personal computer with the following features:

PC with an (*i5-2310*, @ 2.5GHz 2.5GHz) *Intel processor*  
*RAM 6GB running Mint 19.3 and Radeon GPU.*

The following table (Table 1) presents the experiment's parameters of the *DRL*.

**Algorithm 2:** Training process

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1: Input :  $|B|$ ,  $\gamma$ ,  $M$ 
2: Output : Optimal  $Q$  values with trained weights
3: Initialization
4: Randomly initialize all trainable parameters  $\theta$ 
5: Initialize memory  $m$  to capacity  $M$ , batch size  $B$ 
6: Initialize  $Q$ -values randomly for agent
7: Initialize initial state  $s_i$ 
8: Begin
9: Repeat (for each episode) :
10:   Repeat (for each step inside an episode):
11:     Observe current state  $s_t$ 
12:     Choose Action  $a$  and execute it using  $\epsilon$ -greedy
        using Algorithm 1
13:     Take action  $a$ , observe  $r$ ,  $s_{t+1}$  in Eq. (2)
14:     if  $|B| > M$ 
15:       Remove oldest tuple  $t$  in the memory  $m$ 
16:     End
17:     Store tuple  $t = (s_t, a_t, r_t, s_{t+1})$  to memory  $m$ 
18:     Randomly batch_size  $B$  experiences
19:     Update network  $\theta$  using Eq. (5)
20:     Set  $\theta^* = \theta$ 
21:      $s \leftarrow s_{t+1}$ 
22:   Until  $s$  is terminal
23: Until The last episode ends;
24: End

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TABLE I. HYPER-PARAMETERS OF THE AGENT.

| Parameters               | Descriptions                         | Value  |
|--------------------------|--------------------------------------|--------|
| Memory size              | Size of Memory                       | 30000  |
| Batch_size               | Size of batch sampling               | 50     |
| Epoch                    | Training step size                   | 200    |
| Simulation time          | Duration of simulation               | 3000s  |
| Optimizer                | Optimization algorithm               | Adam   |
| Learning rate $\alpha$   | Step size in loss function           | 0.0001 |
| Discount factor $\gamma$ | Weight that multiplies future reward | 0.99   |

The traffic generation in a simulated intersection is considered as a critical step because it can have a significant impact on the global performances. To maintain a high level of reality, the traffic in each episode will be created using an arbitrary distribution [7], however all generated vehicles are identical. An example of such generation is shown in Fig. 3.

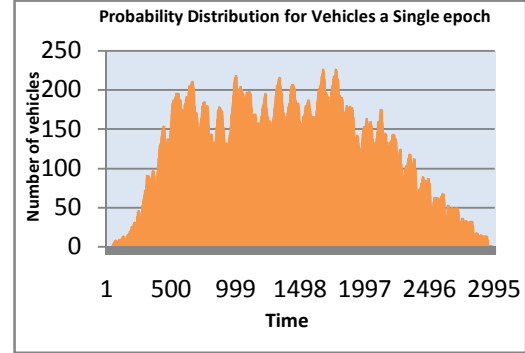


Fig. 3. Traffic signal generation distribution over one epoch.

Fig. 4. displays the training performances, in terms of the four adopted metrics.

We think that it seems very normal to obtain these poor performances towards the beginning of training process because the agent start by executing exploratory actions.

For the evaluation of the performances, we compared the proposed approach with *DQTSCA*, the *fixed* and *TSTMA* (It prioritizes the lanes with the most traffic). All algorithms are assessed in the same environment setting. The Fig. 5 shows that the proposed approach achieves the best performances in terms of all metrics and it seems more stable throughout all the simulation time. Another key benefit of our design approach is that it decreases the number of states to the controlled lanes for a multi-lane traffic intersection, thereby lightening the computing load.

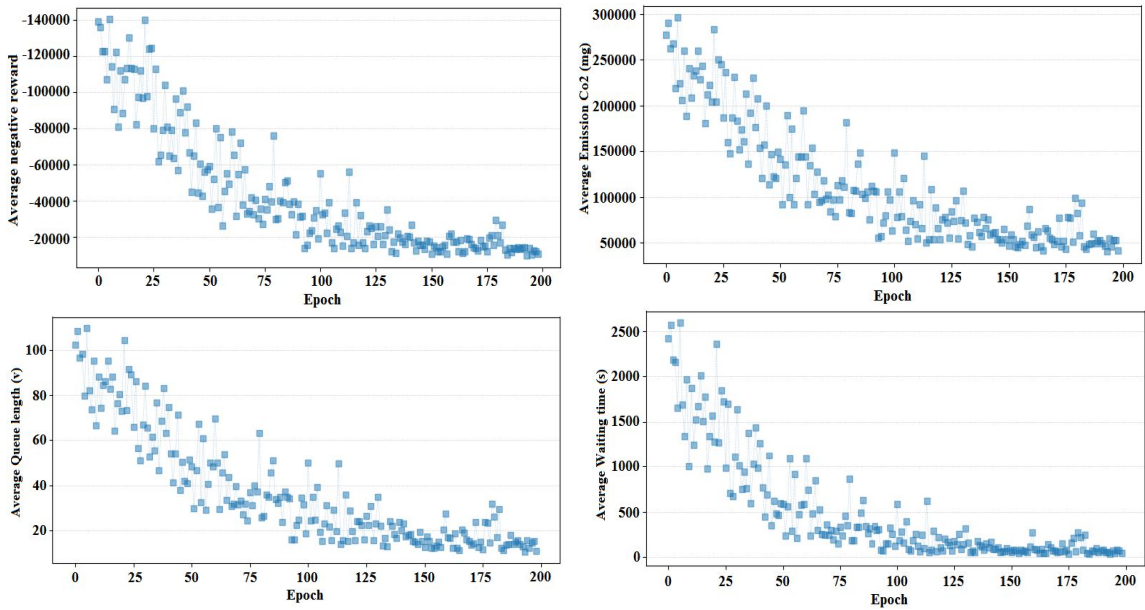


Fig. 4. The training results in terms of cumulative reward, AWT, AEC, and AQL.

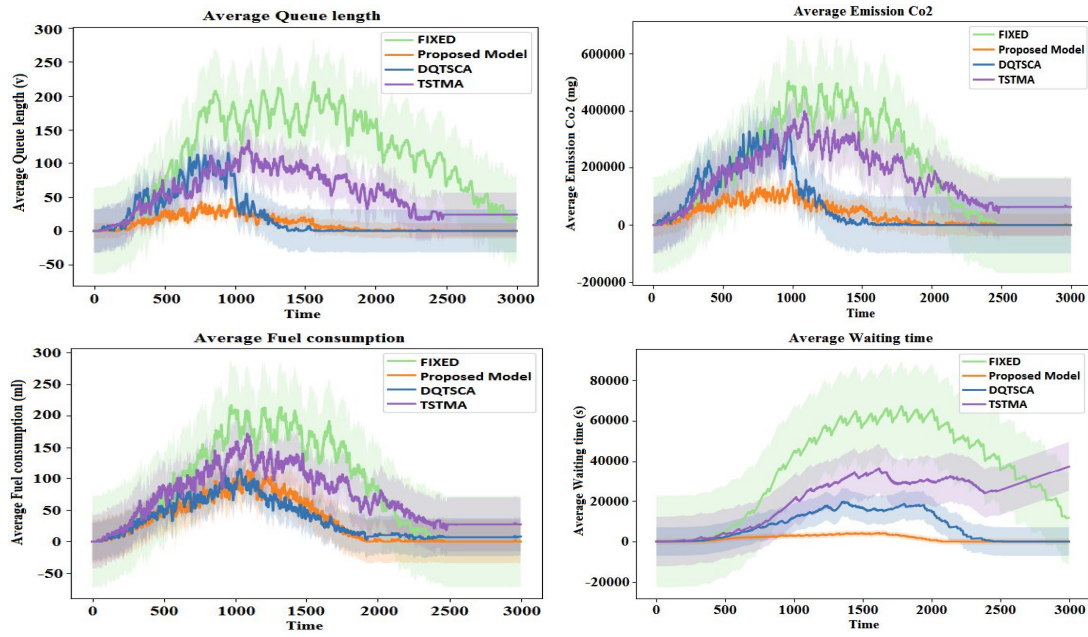


Fig. 5. Plots of performance comparisons of all adopted metrics.

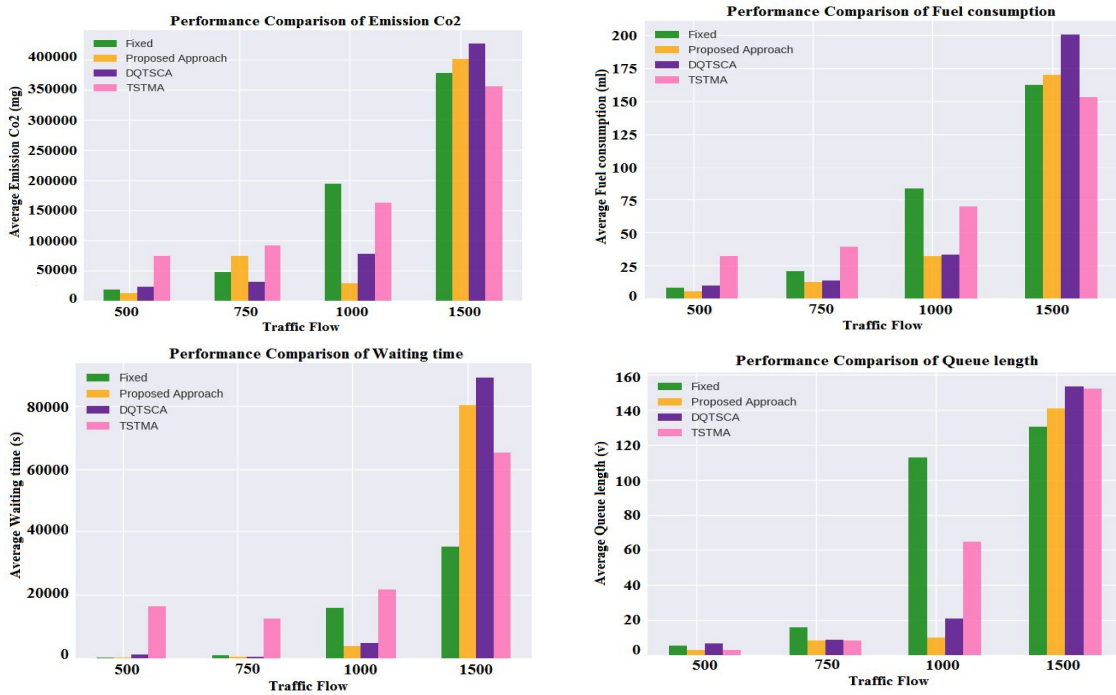


Fig. 6. Performance comparison on average of all adopted metrics for different traffic flow.

We calculate average values over several types of traffic flow (500, 750, 1000, 1500 vehicles per hour) from these experiments as our results, which are shown in Fig. 6. Note these both *DRL* control approaches (proposed approach and *DQTSCA*) were trained in a traffic system with 1000 vehicles per hour traffic flow. It is worth noting our approach to efficient performance on all aspects, except on the traffic flow of 1500 vehicles per hour. There are no notable differences in all measures between these methods (*fixed*, *DQTSCA*) due to the intersection's acceptable capacity at low vehicle production rates. However, when the number of vehicles degrades with increasing traffic production (1000 vehicles per hour), compared to other control algorithms, our approach outperforms them in all metrics. In the majority of cases, especially in high traffic flow (1500 vehicles per

hour), our control strategy provides more reliable and efficient control. Traffic congestion is to blame for this poor performance. Perceiving all results, Table 2 below shows the percentage improvement in proposed approach adjusted all metrics compared to other approaches.

TABLE II. THE PERCENTAGE REDUCED OF OUR APPROACH COMPARED TO OTHER APPROACHES.

| Metrics | Fixed  | DQTSCA | TSTMA  |
|---------|--------|--------|--------|
| AWT     | 30.01% | 17.32% | 40.08% |
| AQL     | 35.32% | 16.02% | 20.54% |
| AFC     | 41.57% | 13.84% | 45.88% |
| AEC     | 16.15% | 12.65% | 39.79% |

## VI. CONCLUSION

In this work, we have tackled the *ATSC* problem. While proposing a new well-designed *DRL* approach. The aim is to reduce traffic congestion in an isolated intersection. The proposed approach can extract valuable information from raw traffic flow in real time automatically. These data constitute the approach's inputs through which our *DDQN* learns the optimal traffic signal control policy. Adopting experiential and target network replay mechanisms, the simulation results showed that our proposition minimizes considerably *AWT*, *AQL*, *AFC* and *AEC* compared to *DQTSCA*, the fixed model and *TSTMA*.

Analyzing the limits of our contribution will allow us to highlight some crucial future directions. Thus, we will continue to work on this problematic, notably to expand the proposed approach to be more applicable in the larger-scale urban traffic network. In this way, it can be notified that this research work deals with only a simplified case that is an isolated intersection, whereas the reality is much more complicated. Thus, as a perspective of this study, we propose to ameliorate the proposed approach to be valid for multiple intersections issues.

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